

DAILY TASK-2

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

= There are 3 types SMPS:-

- i. Desktop Computer, SMPS efficiently converts AC power into the multiple DC voltages needed by computer components.
- ii. LED Television, provides stable and energy efficient power for the TV's electronic circuits.
- iii. Laptop Charger, it converts high-voltage AC power into the low-voltage DC power required to charge the laptops.

- There are 2 devices which use UPS

- i. Desktop Computer, it provides backup power during outages, preventing sudden shutdown and data loss.
- ii. Wi-Fi routers, it keeps the internet connection running during power cuts for uninterrupted communication.

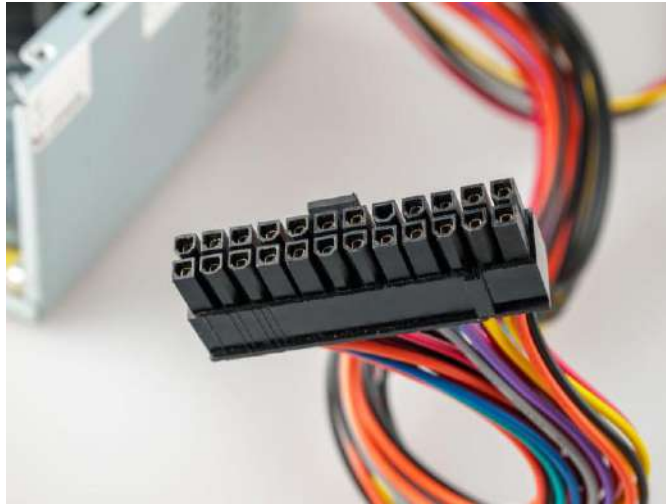
2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.

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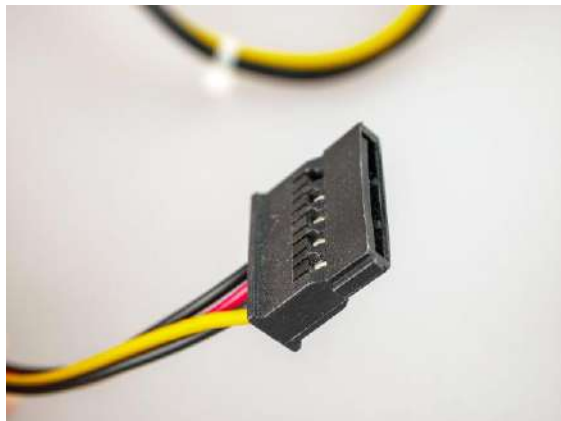
3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

=i. ATX 24-Pin Motherboard connector.



it power Motherboard, RAM, Chipset its safety tips is turn off power and release locking clip before removal.

ii. SATA Power Connector



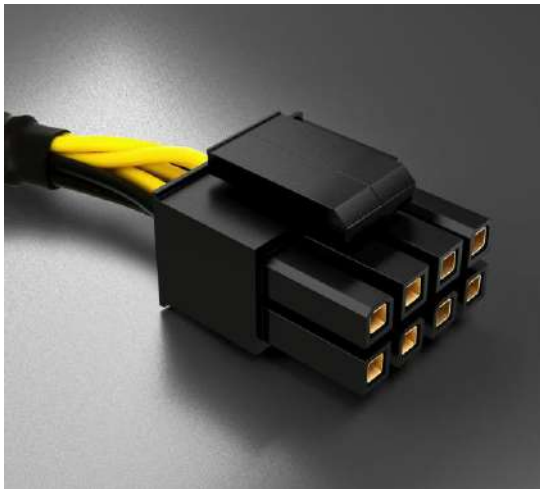
it power SSDs, HDDs, its safety tip is it align the L-shaped key correctly.

iii. Molex Connector



It power legacy drives, fans, etc. its safety tip is pull from the connector body, not the wires.

iv. PCIe 6/8- Pin



it power Graphics card its safety tip is it ensure a secure click and use the correct cable

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

= - Steps

- i. connect the 24-pin ATX cable from SMPS to motherboard.
- ii. connect the 8-pin CPU power cable to the CPU power socket on the motherboard.
- iii. connect the PCIe 6/8-pin cable to the graphics card
- iv. connect SATA power cable to the SSD/HDD storage drives.
- v. check that all cable are firmly connected.

vi. plug in the power cord, turn on the SMPS, and start the pc.